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The Transportation Revolution and the Erosion of the Walking City

Between 1815 and 1875, America's largest cities underwent a dramatic spatial change. The introduction of the steam ferry, the omnibus, the commuter railroad, the horsecar, the elevated railroad, and the cable car gave additional impetus to an exodus that would turn cities "inside out" and inaugurate a new pattern of suburban affluence and center despair. The result was hailed as the inevitable outcome of the desirable segregation of commercial from residential areas and of the disadvantaged from the more comfortable. Frederick Law Olmsted wrote that the city, no less than the private home, had to be divided into various segments that could perform specialized functions: "If a house to be used for many different purposes must have many rooms and passages of various dimensions variously lighted and furnished, not less must such a metropolis be specifically adapted at different points to different needs." Olmsted was writing soon after the Civil War when urban concentrations had generated enough people with enough wealth to provide the market demand for large numbers of private houses near major urban centers.¹

The Appendix provides a quantitative analysis of the beginnings of this change. Essentially, it demonstrates that enormous growth to metropolitan size was accompanied by rapid population growth on the periphery, by a leveling of the density curve, by an absolute loss of population at the center, and by an increase in the average journey to work, as well as by a rise in the socioeconomic status of suburban residents. This shift was not sudden, but it was no less profound for its gradual character. Indeed, the phenomenon was one of the most important in the history of society, for it represented the most fundamental realignment of urban structure in the 4,500-year past of cities on this planet.²

In the most populous metropolitan areas, this shift in residential status

between periphery and core began before the Civil War. In Boston, Harrison Gray Otis, a member of an upper-class family, built a considerable fortune by shaping the expansion of the walking city. In 1795 he formed the Mount Vernon Proprietors and several years later purchased eighteen vacant acres on Beacon Street for \$18,500. Privy to the decision to locate the new Massachusetts State House on adjacent property, Otis promptly developed his land as housing for Boston's elite. As Walter M. Whitehill has written, "This venture . . . was the largest land transaction ever to have been undertaken in Boston at that time, for it involved a sudden change in the character of an entire region."³

Outside of Boston, the most dramatic changes took place in the unannexed villages of Cambridge and Somerville. According to Henry Binford they developed three important suburban characteristics between 1800 and 1850: a set of clear municipal priorities, a preference for residential over commercial expansion, and a stubbornness to remain politically independent from Boston. As the metropolitan population swelled in the 1830s and 1840s, a handful of young, wealthy residents of old Cambridge and Cambridge Port pioneered local commuting. Called "transitional commuters" by Binford, these early suburban residents differed from later commuters because they did not use public transportation and because their original residences were located on the periphery, not in Boston.

In New York City, where extraordinary growth pushed the population of the city and adjacent Brooklyn well past the million mark by 1860, the "genteel" population moved northward from City Hall, especially to the high ground in the middle of the island a few blocks on either side of Fifth Avenue. Building substantial homes along King and Charlton streets or around small parks such as Washington, Gramercy, and St. John's, they brought their elite institutions—the Union Club, the First Presbyterian Church, Grace Church, and Columbia College—with them to the edge of the city.

Change was so rapid in Gotham (Washington Irving's term for New York City) that residents were completely baffled. Phillip Hone, who retired at an early age in 1821 to devote his time to social and public pursuits, including a stint as mayor, complained of being forced out of his comfortable and prestigious home across the street from City Hall by offers for his land which he could not refuse, from people who intended the property for commercial use:

Old downtown burgomasters, who have fixed to one spot all their lives, will be seen during the next summer in flocks, marching reluctantly north to pitch their tents in places which, in their time, were orchards, corn-

fields, or morasses a pretty smart distance from town, a journey to which was formerly an affair of some moment, and required preparation beforehand, but which constitute at this time the most fashionable quarter of new York.⁴

A house at 29 East 4th Street, now a recognized landmark, illustrates the trend. Built in 1832 as part of a speculator's row on a luxury "uptown" block that was still adjacent to farms, the four-story brick structure was bought three years later by Seabury Tredwell, a prosperous hardware merchant and descendant of Samuel Seabury, the first bishop of the Episcopal Church in New York. The important selling features of the house included an elaborate doorway that combined the best of the Federal and Greek Revival styles, a 4,000-gallon cistern in the rear garden that antedated the Croton Aqueduct water system, interior Greek and Ionic columns separated by polished mahogany sliding doors, and double parlors that ran the full length of the building. Although unusual because various Tredwells lived in it from 1835 until 1933, the home in an agricultural area represented the shift of affluent, bourgeois families from old New York to the developing periphery.

As has usually been the case in the United States, the distribution of population was governed primarily by the desire of property owners and builders to enhance their investments by attracting the wealthy and by excluding the poor. Samuel Ruggles, an 1831 purchaser of the 22-acre Gramercy Farm between 19th Street on the south, 23rd Street on the north, the Bloomingdale Road (now Broadway) on the west, and Second Avenue on the east, was particularly adept at developing high-status real estate. Quickly subdividing the farm into 108 city lots, Ruggles then transformed the equivalent of 42 lots into a private park, 520 by 184 feet. The deed read as follows:

Samuel Ruggles proposes to devote and appropriate the said 42 lots of land to the formation and establishment of an ornamental private Square or Park, with carriageways and footwalks at the south-western and north-eastern ends thereof for the use, benefit and enjoyment of the owners and occupants of sixty-six surrounding lots of land belonging to the said Samuel B. Ruggles and with a view to enhance the value thereof.

The value of the land was enhanced as the park—admission to which has traditionally been by a special key available only to the chosen—became the center of a wealthy neighborhood and gave an uptown push to the movement of the affluent away from lower Manhattan. By the time of the Mexican War, Gramercy Park had become a bastion of correct society. Dr. Valentine Mott, reportedly "as great a surgeon as Wellington was a soldier," lived at No. 1; Amos Pinchot, noted politician and former governor of Pennsylvania, lived at No. 2. Numbers 3 and 4,

TABLE 2-1
Occupational Distribution, District of Penn (South Penn)
Philadelphia County, Pennsylvania, 1850^a

<i>New Middle Class</i> ^b	12.5%	<i>Artisans</i>	49.3%	<i>Unskilled</i>	38.2%
Storekeepers	4	Carpenters	13	Laborers	33
Merchants	4	Bricklayers	8	Carters	13
Hotelkeepers	3	Shoemakers	8	Miscellaneous	12
Tobacconists	2	Brickmakers	4	<i>Total</i>	58
Clerks	2	Cabinetmakers	4	<i>Percentage</i>	38.2
Accountant	1	Tailors	4		
Horse Doctor	1	Machinists	3		
Manufacturer	1	Dyers	3		
Salesman	1	Blacksmiths	3		
<i>Total</i>	19	Jewelers	3		
<i>Percentage</i>	12.5	Hatters	2		
		Painters	2		
		Foremen	2		
		Farmers	2		
		Sashmakers	2		
		Weavers	2		
		Misc	10		
		<i>Total</i>	75	<i>Total Sample</i>	160
		<i>Percentage</i>	49.3	<i>Unclassified</i>	8

^aUsing a systematic random sampling procedure, 10 percent of the total number of households in the district were sampled. If the occupation of more than one family member was given, that of the individual listed first was recorded.

^bThe classifications are those suggested by Sam Bass Warner in *The Private City: Philadelphia in Three Periods of Its Growth* (Philadelphia: University of Pennsylvania Press, 1968), 64–65. Professor Herbert Gutman has advised me regarding the skills of certain of these occupations.

SOURCE: U.S. Bureau of the Census, *Population Schedules of the Seventh Census of the United States, 1850. Pennsylvania*.

considered the most beautiful houses on the park, belonged to Mayor James Harper of the publishing house. Vincent Astor at No. 5, novelist Henrietta B. Haines at No. 10, composer and patron of the arts Samuel L. M. Barlow at No. 11, former Secretary of the Treasury Albert Gallatin at No. 24, and the Steinway piano family at No. 26 were among the early notables.⁵

Residential development in Philadelphia was similar to that of its larger rival. In 1850 the City of Brotherly Love counted about 565,000 inhabitants, its built-up area extended over about ten square miles, and its most important axis of growth was out Ridge Avenue to the northwest. Socioeconomic changes in the Penn District between 1850 and 1860, when that area was directly in the path of advancing settlement (TABLES 2-1 and 2-2) suggest the increasing status of peripheral residence. Arti-

TABLE 2-2
Occupational Distribution, District of Penn (Ward 20)
City of Philadelphia, Pennsylvania, 1860^a

<i>New Middle Class</i>	26.9%	<i>Artisans</i>	48.4%	<i>Unskilled</i>	24.7%
Clerks	31	Shoemakers	36	Laborers	69
Merchants	29	Carpenters	33	Car Drivers	7
Grocers	24	Tailors	19	Dressmakers	5
Manufacturers	14	Blacksmiths	15	Conductors	4
Physicians	5	Machinists	10	Paper Hangers	4
Contractors	4	Brickmakers	9	Carters	4
Realtors	4	Butchers	9	Coopers	3
Teachers	4	Moulders	7	Gasworks makers	3
Hotelkeepers	4	Stone masons	7	Tin workers	3
Attorneys	3	Cabinetmakers	7	Domestics	2
Druggists	3	Painters	7	Hatters	2
Clergymen	3	Coachmakers	6	Tanners	2
Chemists	2	Printers	6	Gardeners	2
Gentlemen	2	Jewelers	5	Morocco dressers	2
Tobacconists	2	Plasterers	5	Miscellaneous	16
Millers	2	Saddlers	5	<i>Total</i>	128
Traveling Agents	2	Bakers	4	<i>Percentage</i>	24.7
One Each	5	Ropemakers	4		
<i>Total</i>	141	Brewers	4		
<i>Percentage</i>	26.9	Stone cutters	4		
		Weavers	4		
		Wheelwrights	4		
		Three Each	12		
		Two Each	18		
		One Each	13		
		<i>Total</i>	255	<i>Total Sample</i>	524
		<i>Percentage</i>	48.4	<i>Unclassified</i>	33

^aBoth the methodology and the classification system are the same as those employed in TABLE 5.

SOURCE: U.S. Bureau of the Census, *Population Schedules of the Eighth Census of the United States, 1860. Pennsylvania.*

sans and unskilled laborers who walked to work in the adjacent industrial suburbs of Spring Garden and Kensington were the dominant group in both years.⁶ But as population soared in formerly empty areas, the proportion of merchants, manufacturers, physicians, and grocers, the "new middle class," in Sam Bass Warner's phrase, approximately doubled.⁷ After the Civil War, the number of commuters to the center city continued to increase as businessman opted for "all the beauties of the country, within an easy and cheap communication with the city."⁸

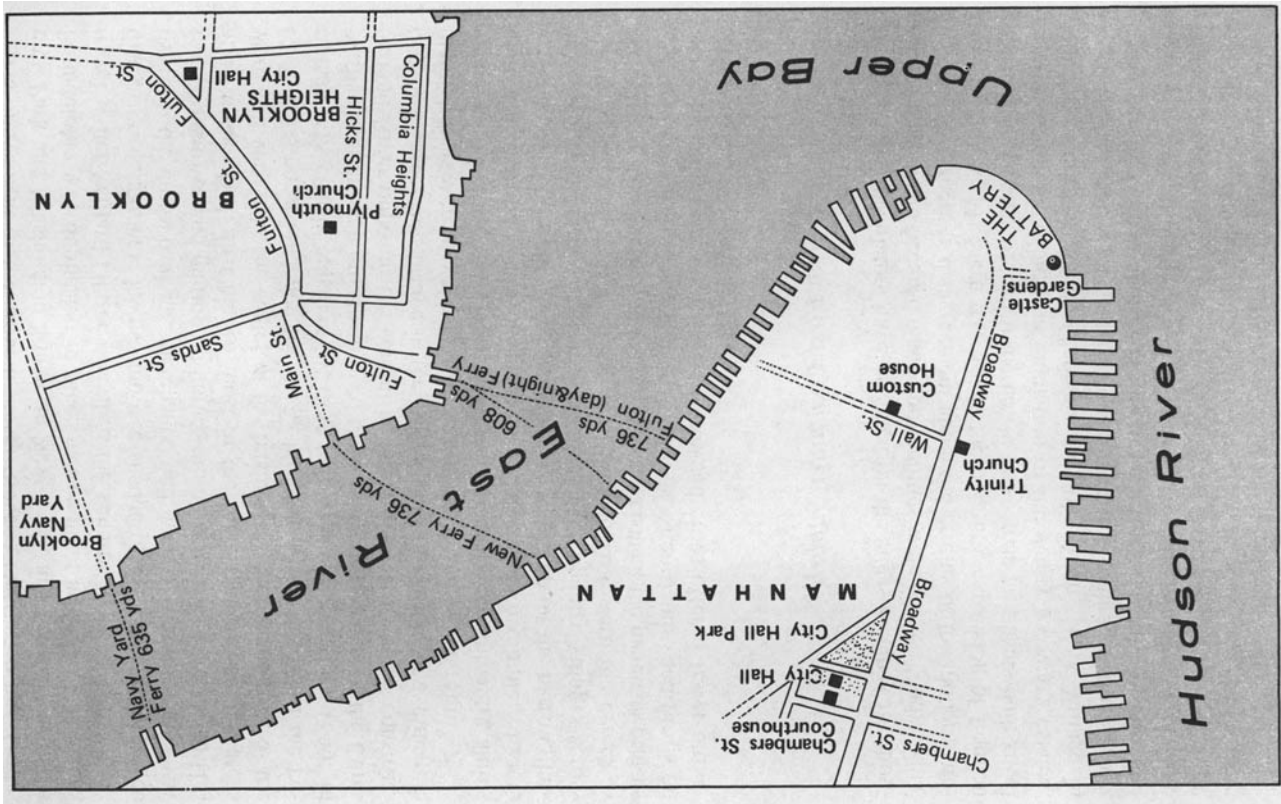
The movement of the affluent toward Beacon Hill in Boston, Gramercy Park and Washington Square in New York, and Germantown near Philadelphia was duplicated in other metropolitan areas. In Cincinnati, travel writer Willard Glazer described the suburbs in 1883 as a "Paradise of grass, gardens, lawns, and tree-shaded roads." In San Francisco by 1860 the city's bankers, merchants, and doctors were moving away from downtown and putting up homes on the heights of Fern (Nob) Hill and Russian Hill. In Chicago most of the high-grade residential areas were still very near the center of the business district at the time of the Civil War, but a tendency for the fashionable to move toward the periphery was clearly apparent by 1873 and was widespread by 1899. In Nashville prominent citizens moved to Edgefield, lying east of the Cumberland River opposite the central business district, after the construction of a suspension bridge in the 1850s. And in Buffalo new elite residential areas were developed at the outer edges soon after 1860.⁹

In smaller cities and towns, suburbs remained predominantly slums until well into the twentieth century. In Calgary, for example, clusters of cheap dwellings and "shack towns" were put up on the narrow flats below the ridge as late as World War I. And in country villages across North America, the best streets were often those toward the center as late as 1970. But by 1875 in the major urban centers, the merchant princes and millionaires were searching for hilltops, shore lands, and farms on which to build substantial estates; crowded cities offered fewer attractions with every passing year.¹⁰

The First Commuter Suburb

Because land speculators have operated on the edges of cities for thousands of years and because many communities were labeled "suburban" before 1800, there is disagreement about the origins of the modern suburb. Robert A. M. Stern traces it to the booming expansion of London under King George III, when the newly prosperous merchants built small houses in remote villages in emulation of the gentry's country estates. Other possibilities include Clapham and John Nash's Park Village in London, Cambridge outside Boston, Greenwich Village north of New York, Spring Garden and the Northern Liberties near Philadelphia, and New Brighton on Staten Island. None of these communities offered the number of commuters, the easy access to a large city, and the bucolic atmosphere of Brooklyn Heights, which grew up across the harbor from lower Manhattan in the early decades of the nineteenth century.¹¹

The Canarsie Indians, calling the high, sandy bank that is now Brook-



lyn Heights “Ihpetonga,” lived there for centuries in communal houses that were hundreds of feet long before the Dutch arrived in 1636. But unlike more commercial New Amsterdam, which had been founded across the East River in 1624, Brooklyn remained essentially agricultural until 1800, and its economic ties to the metropolitan entrepôt were slight. As late as 1810, it was occupied mostly by farms, and its population was less than five thousand (TABLE 2-3).¹²

TABLE 2-3
Population of New York City, Brooklyn, and Flatlands,
by Decades, 1790–1890

Year	New York City	Brooklyn	Flatlands
1790	33,131	1,603	423
1800	60,489	2,378	493
1810	96,373	4,402	517
1820	123,706	7,175	512
1830	202,589	15,394	596
1840	312,710	36,233	810
1850	515,547	96,838	1,155
1860	813,669	266,661	1,652
1870	942,292	396,099	2,286
1880	1,206,299	566,663	3,127
1890	2,515,301	806,343	4,075

^aThe purpose of this table is to demonstrate the effect of geography upon community growth. Flatlands and Brooklyn were both nineteenth-century towns within Kings County. Brooklyn, however, was immediately adjacent to lower Manhattan, while Flatlands was on the eastern edge of the county, about eight miles overland from New York.

SOURCE: Ira Rosenwaike, *Population History of New York City* (Syracuse, 1972), especially Tables 4 and 15.

In the next four decades, however, the town of Brooklyn was transformed. Regular steam ferry service to New York City (then consisting only of Manhattan) began in 1814, and one year later the *Brooklyn Star* predicted that the town “must necessarily become a favorite residence for gentlemen of taste and fortune, for merchants and shopkeepers of every description, for artists, artisans, merchants, laborers, and persons of every trade in society.” The accuracy of this prophecy soon became apparent. With its tree-shaded streets, pleasant homes, access to Manhattan, and general middle-class ambiance, Brooklyn attracted those who sought respite from the extraordinary bustle and congestion of Gotham. Walt Whitman, whose office at the *Brooklyn Eagle* overlooked the Fulton Ferry slip, frequently commented on the suburb’s phenomenal growth.

"There," he said, "men of moderate means may find homes at a moderate rent, whereas in New York City there is no median between a palatial mansion and a dilapidated hovel." Writing often of the river, the fog and the ferries, Whitman captured the commuting character of the place he called "Brooklyn the Beautiful."¹³

In the morning there is one incessant stream of people—employed in New York on business—tending toward the ferry. This rush commences soon after six o'clock. . . . It is highly edifying to see the phrenzy exhibited by certain portions of the younger gentlemen, a few rods from the landing, when the bell strikes . . . they rush forward as if for dear life, and woe to the fat woman or unwieldy person of any kind, who stands in their way.

Additional ferry lines soon expanded the commuting possibilities. In 1836, the South Ferry began regular runs from Whitehall Street to Atlantic Street in Brooklyn, and in 1846 the Hamilton Ferry began connecting Hamilton Avenue with the battery in Manhattan. By 1854 the Union Ferry Company had consolidated a dozen competing lines and was providing 1,250 crossings daily at a standard one-way fare of two cents.¹⁴ Williamsburg, for example, was served by six steam ferries leaving Peck Slip every ten minutes and Grand Street every five minutes during the working day. Its population shot from 3,000 in 1835 to 40,000 in 1852. By 1860 the various East River ferries were carrying 32,845,000 passengers per year (or about 100,000 per working day), and by 1870 the patronage had increased to 50 million per year. Indeed, the folk wisdom of the day held that when there was fog in the harbor, half the business population of Manhattan would be late for work. Actually, only 40 percent of Brooklyn wage earners worked in Manhattan in 1860, but they tended to be above average in wealth and position. As the ferry districts became distinct from the more rural areas of the rest of Kings County, so the political form began to change. In 1834 Brooklyn leaders won city status from the state legislature over the opposition of Manhattan representatives. By that time suburban landowners and speculators were anxious to subdivide their farms into city lots, and they perceived that a stronger government would provide the streets and services that would aid the rapid development of the periphery. But high taxes, hopefully, would not be necessary, as real-estate investors were quick to emphasize. In 1854, when James Cole and Sons advertised 450 lots near Carroll Park, the firm noted optimistically:

The distance of these lots from the Merchants Exchange, New York, is the same as to Tenth Street in that city, but requiring only half the time to pass, on account of the rapid transit over the long ferries between the two

cities . . . ; the consolidation of Brooklyn and Williamsburgh will have a tendency to reduce and permanently fix the tax on property which is at present far less than the taxes of New York, while the actual cost of real estate in Brooklyn does not exceed one-fifth of what it is in corresponding parts of that city.¹⁵

Whether the attraction was easy access, pleasant surroundings, cheap land, or low taxes, the suburb was growing at a faster rate than the city by 1800, and in almost every decade until the Civil War its population approximately doubled. One wag noted that Brooklyn “sold nature wholesale” to real-estate developers, for sale to homeowners at retail. In 1853, at single sales, 360 lots were sold in Bushwick, 150 in Fort Hamilton, 600 in Brooklyn proper, and 950 in Newton. Even on the farthest outskirts of the built-up region, the influx of middle-class families soon became apparent. In 1850, for example, the little village of Bedford (now part of Bedford-Stuyvesant, the largest black neighborhood in the Northeast), was made up of a variety of occupations, reflecting its essentially rural orientation. By 1880, however, when it had become part of the expanding metropolis, very few laborers remained, and the farmers had disappeared.¹⁶ Newspaper advertisements offered a home in the suburbs, no farther from the heart of Manhattan than many tenements, for 10 to 40 percent down and payments spread over three to five years. Developers were not only building elegant structures for the wealthy featuring “clean sea breezes and a glorious view of greater New York and its harbor,” but inexpensive dwellings for the middle classes.¹⁷

Far from seeing the growth of Brooklyn as an advantage, New York newspapers, politicians, and land developers by 1850 brooded over the intense competition and expressed concern over “the desertion of the city by its men of wealth.” The Association for the Improvement of the Condition of the Poor moaned that “many of the rich and prosperous are removing from the city, while the poor are pressing in,” and as early as 1840 Alderman Daniel F. Tieman was opposing the improvement of the ferry service (an operation controlled and regulated by the City of New York by legislative fiat to the continuing chagrin of Brooklyn) on the assumption that even more people might thereby be induced to forsake Gotham for the more bucolic atmosphere across the river. Instead, Tieman and his supporters suggested a fare rise of more than 700 percent (from 3 to 25 cents).¹⁸ The *New York Tribune* summed up the indictment against suburban competition on January 21, 1847:

Property is continually tending from our city to escape the oppressiveness of our taxation; many who have made fortunes here carrying them away to be expended and enjoyed, while thousands who continue to do

business here, reside, and are taxed elsewhere on the same amount. Thus, while every suburb of New York is rapidly growing, and villages twenty and thirty miles distant are sustained by incomes earned here and expended there, our City has no equivalent rapidity of growth, and unimproved property here is often unsalable at a nominal price.

As a result of the continuing exodus and spillover from New York, Brooklyn was gradually transformed from a suburb into a major city in its own right, the fourth largest in the country in the latter part of the nineteenth century. By 1890 it counted more than 261,000 foreign-born residents, equal to its entire population only thirty years earlier. With this increase in size, the decay, noise, and fast lifestyle so many had fled Manhattan to avoid followed them across the river. Even though Brooklyn's initial budding was due to a quiet environment easily accessible to the central business district of the world's busiest seaport, its later growth was the result of the development of its own commerce and industry. By 1890 Brooklyn possessed large hat factories, chemical works, foundries and iron mills, candy companies, and coffee and syrup mills. Only Chicago had a larger dressed-meat operation, and no place on earth had larger sugar-refining and grain-depot operations.¹⁹

Hezekiah Beers Pierrepont

Brooklyn's meteoric rise as the premier suburb of the first half of the nineteenth century was mirrored by the career of Hezekiah Beers Pierrepont.²⁰ Born in 1768 and the grandson of one of the founders of Yale College, he struck out on his own at an early age because he disliked Latin and Greek. Working first at the Customs House in New York City and later as a financier, Pierrepont quickly demonstrated an ability to make money. In 1793 he formed a partnership with his cousin, moved to Paris, and began importing provisions into France, where scarcity prevailed as a result of the recent revolution. Successful despite the Reign of Terror (Pierrepont witnessed the execution of Robespierre on July 28, 1794), he subsequently expanded his operations to India and China. His foreign ventures turned sour in 1797, however, when his heavily laden cargo vessel, the *Confederacy*, was lost to pirates. All efforts at recovery failed, and the 29-year-old entrepreneur was bankrupt.²¹

Returning to New York City in 1800, Pierrepont took a more common route to fortune. In 1802 he married Anna Marie Constable, the daughter of merchant William Constable, the largest single landowner in New York State. Receiving half a million acres in Oswego, Jefferson, Lewis, St. Lawrence, and Franklin counties as a wedding present,

Pierrepoint abandoned foreign commerce as too risky and decided to service the more predictable demand for alcohol. For this purpose, he purchased a gin distillery at the foot of Joralemon Street in Brooklyn. Soon thereafter, in 1804 he bought the nearby Benson farm and its spacious residence—Four Chimneys—for his bride.²²

The brewery did not prosper, so Pierrepoint soon turned his energies to land speculation and community boosterism. More than any other individual, he recognized the future prospects of his little village, reasoning correctly that great profits would accrue to those who acted with dispatch. He expanded his local real-estate holdings (the Robert De Bevoise farm and part of the Joris Remsen farm were among his acquisitions) to a total of sixty acres, including eight hundred feet of frontage on the New York harbor, and he became active in local politics.²³ Pierrepoint was a member of the Committee of Fifteen which obtained a village charter for Brooklyn in 1816, and he served thereafter as a trustee. Similarly, he made his influence felt in the opening and widening of roads that would effect his holdings. Unhappy with the first street plan presented to the trustees, Pierrepoint hired his own surveyor, developed an alternative proposal more attractive to his financial interests, and pushed it through to adoption. He then had his property marked off into 25-by-100-foot lots. In thus combining the roles of land speculator and local politician, Pierrepoint was establishing a tradition that would typify residential settlements of all varieties in the United States.

A close friend and financial supporter of Robert Fulton, who developed the first successful steamboat in the hemisphere in 1807, Pierrepoint initially held his lots off the market and bided his time until the improvement of East River transportation would enable him to sell them at handsome prices. In 1814 Fulton began the world's first steam ferry service between Beekman Slip in Manhattan (renamed Fulton Street in 1816 in honor of the inventor who died in 1815) and Fulton Street in Brooklyn. Each of the ferries was large enough to accommodate two hundred passengers and numerous horses and wagons and fast enough to make the crossing in eight minutes. Pierrepoint meanwhile was formulating his plan for a high-class residential area for affluent businessmen who could purchase several contiguous lots and combine them into one prestigious property suitable for a large mansion. In 1820 the "Heights" still boasted only seven houses, but an 1823 advertisement for "Lots on Brooklyn Heights" was a harbinger of change:

Situated directly opposite the southeast part of the city, and being the nearest country retreat, and easiest of access from the center of business that now remains unoccupied; the distance not exceeding an average fifteen to twenty-five minute walk, including the passage of the river; the ground elevated and perfectly healthy at all seasons; as a place of resi-

dence all the advantages of the country with most of the conveniences of the city.

Gentlemen whose business or profession require daily attendance into the city, cannot better, or with less expense, secure the health and comfort of their families than by uniting in such an association.²⁴

As we have seen, the appeal of Brooklyn Heights was as Pierrepont expected. In 1841 of the eighty-four heads of households residing on land purchased from this early developer, thirty-nine worked in "the city," twenty-six of them as merchants. Included among the new suburbanites were William S. Packer, a wealthy, retired fur merchant, and Abiel Abbott Low, a shipping magnate successful in the China trade.²⁵

Hezekiah Pierrepont died in 1838, before he realized his final dream of constructing a public promenade along the edge of the waterfront.²⁶ But his heirs continued the division and sale of the estate lands, sometimes in large parcels intended for further subdivision. And Pierrepont himself did live long enough to see the impeccable dwellings, some of them conventional urban row houses, of Montague, Remsen, and Pierrepont streets, which were as peaceful as New York City was bewildering and noisy, contribute to an image of Brooklyn that was attractive to the middle class.

Just as Hezekiah Pierrepont symbolized the first generation of Brooklyn suburban growth, Edwin Clark Litchfield typified the second. With a fortune based on railroad promotion in the Middle West, Litchfield began in 1852 a series of extensive land purchases in Park Slope. The tract he ultimately assembled included the high ground that is now the central portion of Park Slope as well as a broad area of meadow and marsh down near the Gowanus Canal. Litchfield's grand design called for residential development of the upper section and commercial and industrial use of his properties near the canal, but his first step was to construct an elaborate Italianate residence—Grace Hill—to set an aristocratic tone for the neighborhood. The project was spectacularly successful; the area became among the most desirable in the metropolitan region, and Litchfield's home became the Brooklyn headquarters of the New York City Department of Parks and Recreation. Part of his yard was incorporated into Prospect Park, the most noble of Frederick Law Olmsted's many creations.

Other Ferry Suburbs

Brooklyn was the first and most important of the modern "ferry suburbs," but water commutation was also significant elsewhere. Before the

Holland (1927) and Lincoln (1937) tunnels were completed, there were a score of ferry lines across the Hudson River to New Jersey. In 1821 Colonel John Stevens, inventor of the screw propeller and the owner of the entire island of "Hobuck," established regular steam service to Hoboken, which soon became a well-known pleasure spot.²⁷ John Jacob Astor built a summer house there in 1829, and soon thereafter the Weehawken Palisades became the site of elaborate estates of wealthy New Yorkers. Eventually, thousands of ordinary pleasure-seekers crossed the Hudson every weekend in the summer months to enjoy its delights. By the early 1850s, ferries shuttled between Jersey City and Manhattan every fifteen minutes, and by 1870 daily commuting to Gotham had become an established feature of middle-class life in Hudson County, with the Jersey City ferry alone bringing more than three thousand passengers per day into the city.²⁸

No other American harbor generated nearly as much water-borne commutation as that of New York, but daily ferry riding was also a common pre-Civil War experience between Philadelphia and Camden, Newport and Cincinnati, and Pittsburgh and Allegheny City. In San Francisco wealthy merchants began building residences on the opposite side of the bay in Oakland and Alameda in the 1850s, while in Boston the two-cent and five-minute all-water journey to Noodle's Island and East Boston was daily attracting more than a thousand riders.²⁹

The Origin of Public Transportation Systems—The Omnibus

The application of steam power to water travel had a substantial impact on only a limited number of cities. But many large communities were confronted with growth that was so substantial that it outmoded pedestrian movement as a workable basis for organizing urban space. Buildings became larger and streets more crowded, so that congestion could be relieved only by making more territory accessible to city residents. The obvious solution was the establishment of ground transportation networks in urban areas. Prior to 1825 no city anywhere possessed a mass-transit system—which may be defined as operation along a fixed route, according to an established schedule, for a single fare.³⁰ Horse-drawn carriages for hire—sometimes known as taxis or hackneys—represented the public mode for short trips, while stagecoaches served a similar function for more distant journeys. The first transit system came about almost by accident in 1826, when a retired French Army officer purchased some public baths several miles outside Nantes in western France. To improve accessibility to his business, Baudry initiated a short

stage line between the center of Nantes and the resort. The coaches were popular almost immediately, but Baudry noticed that most of the riders were getting off at intermediate pick-up points rather than continuing on to the baths. Capitalizing on his discovery, Baudry promptly expanded the number and the routes of his short-haul stages, which he dubbed "omnibuses" after the name over the door of a hatmaker named Omnes. The experiment was successful, and on January 30, 1828, the first bus in Paris traveled from the Madeleine to the Bastille. By 1832 Baudry's rudimentary transit system had been copied in Bordeaux, Lyons, and London. Essentially, the omnibus combined the functions of the hackney and the stagecoach.³¹

Abraham Brower introduced omnibus service to North America with operation along Broadway in New York in 1829; enterprising opportunists then took the idea to Philadelphia in 1831, to Boston in 1835, and to Baltimore in 1844. The typical pattern was for a city government to grant a private company—usually a small businessman already in the livery or freight business—an exclusive franchise to operate coaches along an existing street. In return, the company agreed to maintain certain minimum standards of service. Over the years, the cost of a single ride averaged about ten cents, although many firms sought to develop commuter service to new residential areas by offering annual season tickets that permitted an unlimited number of rides at an average expense of four cents per day.³²

By midcentury, the omnibuses had become a big business and had added noticeably to the traffic congestion that had already become a standard feature of urban life. New York City alone counted eighty licensed coaches in 1833; 108 in 1837; 255 in 1846; 425 in 1850; and 683 in 1853, when twenty-two separate firms were competing for the trade, and when average waiting time on some corners was less than two minutes. At one intersection on lower Broadway coaches reputedly passed on an average of every fifteen seconds. So important did the service become to Gotham's economy that when a fire consumed most of the equipment of the busiest West Side line, downtown merchants and city officials stumbled over themselves in helping to get the omnibuses back on the streets. And when the little boxes on wheels appeared in Baltimore in 1844, the *Sun* enthusiastically predicted that they would enable "persons to reside at a distance from their places of business in more healthy locations without loss of time and fatigue in walking."³³

Although the omnibus represented an obvious improvement over walking, the unpadded benches, poor ventilation, and rude, bad-tempered drivers did cause fatigue and exposed the limitations of ground level transportation. The heavy, twelve-passenger vehicles were se-

verely hampered by the condition of city streets, which were paved with uneven cobblestones at best and deeply rutted at worst. Riding in such circumstances was an emotional and physical trial even for the hearty. As the *New York Herald* complained in 1864: "Modern martyrdom may be succinctly described as riding in a New York omnibus."³⁴ As if such ordeals were not enough, omnibus travel speed was invariably less than five miles per hour and often no quicker than the pace of a brisk pedestrian. Not surprisingly, even in New York City, which had the most extensive omnibus network, only about 25,000 persons, or about one resident in thirty, used this form of public transportation on a daily basis in 1850. The significance of this primitive system, however, was that it encouraged an influential minority of urban citizens to develop what Glen Holt has called the "riding habit."³⁵

Steam Railroads

By the middle of the nineteenth century, steam engines with the ability to do the work of a hundred horses were coming into common use. In particular, the steam railroad had a profound influence in reshaping the American city. The railroad was of course an English invention, having been first developed in 1814 by George Stephenson, an unlettered engine-wright in a coal mine. It was his idea that coals could be the power source to transport other coals to the waterside. Exactly half a century later, in *Our Mutual Friend*, Dickens described oozing London's edge, "where Kent and Surrey meet, and there the railways still bestride the market gardens that will soon die under them." In North America, the first important line to begin laying rails was the Baltimore and Ohio Railroad in 1829.³⁶

The first railroads were designed for long distance rather than local travel. But they sought revenues wherever they could find them and very early on built stations whenever their lines passed through rural villages on the outskirts of the larger cities. In the nation's largest city rudimentary commuter travel by steam railroad began in 1832 and by 1837 the New York and Harlem Railroad offered regular service to 125th Street.³⁷ The achievement led to the construction of a new hotel and to the prophecy of the *New York Herald* that "this and other improvements will make Harlem a fashionable rival to Hoboken, New Brighton and other summer resorts."³⁸ Additions to this same line led to central Westchester County by 1844 and to the New York *Tribune* prediction, "The line of this road will be nearly one continuous village as far as White Plains by 1860." Meanwhile, the New York and New Haven Railroad along Long

Island Sound reached New Haven in 1843, and the Harlem River line toward Albany reached Peekskill in 1849. Along these tracks population grew by more than 50 percent in the first decade after initial construction, as real-estate developments sprang up in Rye, Tarrytown, and New Rochelle.³⁹ Between 1850 and 1860, the population of Westchester County as a whole grew by 75 percent, and that portion nearest the city more than doubled. As early as 1855, English observer W. E. Baxter noted that suburban villas were “springing up like mushrooms on spots which five years ago were part of the dense and tangled forest; and the value of property everywhere, but especially along the various lines of railroad, has increased in a ratio almost incredible. Small fortunes have been made by owners of real estate at Yonkers, and other places on the Hudson River.”⁴⁰

Because annual commutation rates (forty-five dollars to Bronxville in 1853) were too high for most wage earners, the railroads helped to foster the image of Westchester as a leafy enclave for the well-to-do. But even before the Civil War, the southernmost stations at Fordham, Morrisania, Tremont, and Mount Vernon were becoming centers of middle-class residence. Morrisania in particular was completely transformed between 1850 and 1865. A small village on the Boston Post Road in the south central part of what is now the Bronx, Morrisania was the scene of large-scale building activity after the opening of a railroad station in the mid-1840s. A particularly important 400-acre subdivision in the 1860s was called Old Morrisania and located near the Gouverneur Morris mansion.

Just as the northbound railroads opened up Westchester County (including what would later become the Bronx), largely replacing the steamboats that had run to Yonkers and Peekskill, so did the Long Island Railroad and the New York and Flushing Railroad enable former Manhattanites to commute from the east. The original purpose of the Long Island Railroad was to link New York City and Boston. Because of “the hills of Connecticut and the many wide and deep New England rivers,” a direct mainland route was not feasible. Railroad promoters believed that the solution was to be found on Long Island, which “followed the general shoreline of Connecticut and which had neither rivers nor hills.” The initial route, which extended “six miles or more north of the south shore, through the unimproved and barren pine-plain area,” terminated at Greenport, where passengers boarded a steamboat to Stonington, Connecticut, and from there a mainland train to Boston. The first run was made on July 27, 1844, with great success. Meanwhile, almost as an afterthought, the LIRR had established commuter service along the route by 1860.⁴¹ The New York and Flushing Railroad, obviously

more interested in short-haul traffic, was unabashedly the instrument of real-estate speculators. By bringing villages in what is now the borough of Queens within one hour of Manhattan in the late 1850s, the railroad led to an influx into Newton, Maspeth, and Flushing.⁴²

Railroad commuting was well established in many other cities before the Civil War. On the other side of the Hudson River from Manhattan, Jersey City was the hub of a combined ferry/railroad route which enabled commuters to traverse the sixteen miles from South Orange to New York City in less than an hour. And during the 1850s, the rail route between Newark and Jersey City was one of the busiest in the world. By 1859 in Philadelphia, more than forty trains were making commuter stops in the northwestern suburb of Germantown, while in Chicago the northern town of Evanston was growing rapidly because of the frequent service of the Chicago and Milwaukee Railroad.⁴³

Boston, however, had a larger proportion of suburban riders than any other community. Along the main line of the Boston and Worcester passengers could travel to Brookline by 1834, and within a decade seven different Hub-based companies were pioneering with such devices as commuter and family fares and free weekend round trips for Bostonians who might be tempted to purchase land in outlying areas. The new price schedules occasionally reduced the price of railroad commutation to that of an omnibus ride within the city of Boston. The tactic worked as railroads and realtors propagandized together in behalf of suburban living.⁴⁴ "Somerville, Medford, and Woburn," read an advertisement for the Boston and Lowell, "present many delightful and healthy locations for a residence, not only for the gentleman of leisure, but the man of business in the city, as the cars pass through these towns often during the day and evening, affording excellent facilities for the communication with Boston." Suburban developers reciprocated by including railroad timetables in the advertisements and reminding potential purchasers that every lot was "within a few minutes walk of the station."⁴⁵

Commuter fares, which were often sold on an annual basis and which ran about sixty-two dollars to Lynn and fifty dollars to Newton in the early 1840s, became particularly important as competition among the railroads increased and prices fell. When some annual suburban round-trip fares fell to as little as thirty dollars per year, the steam railroad became financially competitive with the omnibus.⁴⁶ As a result, there were fifty-nine commuter trains coming into Boston every day from fifteen miles or less (another forty-five came from longer distances) as early as 1849. Places like Dedham, Milton, Quincy, Dorchester, Brighton, Newton, Medford, Melrose, Malden, Winchester, Somerville, and West Cambridge were being labeled by the *Boston Evening Transcript* (April

4, 1855) as railrod suburbs,⁴⁷ and Ralph Waldo Emerson characterized “the readers and thinkers of 1854 as the men on the morning train into the city.” Henry David Thoreau was even swept up in the enthusiasm, noting in his journal that “five times a day I can be whirled to Boston within an hour.”

The enthusiasm of the railroad barons and real-estate developers was not always shared by the general population. Ministers denounced the companies for running trains on Sundays, and farmers regarded trains as unnatural enemies, which shattered their solitude, frightened cows out of giving milk, and blackened the countryside with cinders and soot. Like their contemporary Henry David Thoreau, they objected to whistles and clattering wheels; unlike Thoreau, they occasionally tore up tracks, burned down stations, and caused wrecks by pulling spikes out of the roadbed.⁴⁸

The disruptive steam railroads encountered even more serious difficulties in built-up areas. In New York City in 1839, a locomotive boiler exploded at Fourteenth Street, killing the engineer and injuring twenty passengers. As a result of this and other accidents, fear of the big engines became widespread, and on more than one occasion mobs even tore up tracks on the Bowery. City fathers in both New York and Brooklyn soon banned locomotives from the populous parts of their town (below 14th Street in Manhattan) and required that horses pull the trains into open countryside when steam engines could take over. Similar restrictions in Philadelphia forced the railroads to terminate west of the Schuylkill River.⁴⁹

One way to quell such fears was to miniaturize the engines and to disguise the locomotives to look like passenger cars rather than fearsome monsters. The result was the introduction of so-called “dummy” engines, which were carefully constructed so as to appear harmless. In the long run, however, such efforts at concealment proved unnecessary. As accidents became less frequent, the steam locomotive became an accepted part of the urban landscape. More importantly, the locomotive came to be regarded as a positive force for good, as a means of enabling families to escape the congestion of the crowded city. This “moral influence” of the railroad led supporters of mechanical power to support actively railroad extension. When in 1849 some citizens opposed the extension of the Harlem Railroad south to Chambers Street, the New York *Tribune* countered:

We hope yet to see every part of our City penetrated by railroads, so that nearly every citizen may take a car within two blocks of his store or shop, and be swiftly carried out to his residence amid green fields and waving

forests for a trifle, and the unhealthy packing of ten thousand human beings into three or four blocks of buildings will be gradually overcome.⁵⁰

The Horse Railway

The very rapid adoption of the steam railroad in the United States after 1830 led to the suspicion that placing the omnibus on iron rails and iron wheels would vastly improve service. Initially developed by John Mason on regular railroad tracks between Prince and Fourteenth streets in lower Manhattan in 1832, the horse-drawn streetcar, popularly known as the horsecar, promised to combine the low cost, flexibility, and safety of animal power with the efficiency, smoothness, and all-weather capability of a rail right-of-way. The horsecar was, in short, a blend of the virtues of omnibus and train.

The great expansion of horse-drawn railways came after 1852, when Alphonse Loubat developed a type of grooved rail that lay flush with the pavement. This was an essential improvement because the earliest horsecars had used rails which protruded six inches or more above street level, seriously interfering with coach and wagon traffic and preventing adoption of the innovation. By 1855 the horsecar was forcing the omnibus off the major thoroughfares and onto the secondary routes in New York, and by 1860 the same process was taking place in Baltimore, Philadelphia, Pittsburgh, Chicago, Cincinnati, Montreal, and Boston. The first tracks for street railways in Europe were laid at Birkenhead in 1860 and London in 1861.⁵¹

The great advantage of the horsecar obviously lay in its use of rails, which made possible a much smoother ride at a speed (6–8 mph) almost twice as fast as the omnibus, an important consideration if one wanted to live at a distance from work. Moreover, the reduced friction enabled a single horse to pull a thirty- to forty-passenger vehicle that had more inside room, an easier exit, and more effective brakes than the typical omnibus. All these advantages lowered operating costs, ultimately reducing the average fare for a single ride from fifteen cents on the omnibus to ten cents on the horsecar. The only person whose ride was not noticeably improved was the driver, who sat unprotected from the weather on an open platform. It was thought that if the platform were enclosed, the driver's attention and alertness might be compromised.⁵²

The usual pattern was for tracks to be laid radiating outward from the center of the city in a linear fashion. The horsecar tracks followed the main roads and generally were developed toward the emerging wealthy neighborhoods on the periphery. Indeed, the very availability of quick, efficient mass transportation enhanced property attractiveness, a ten-

dency not lost on land speculators. Subdividers nestled their plots close to the tracks and advertised their proximity to the new convenience. In Boston, where developer George Brastow of Somerville was the leading figure in forming the Middlesex Horse Railroad Company, the horsecar lines tended to extend real-estate development in the same direction as the earlier omnibus. In St. Louis mass transportation extended the built-up area to the northeast, while the equally attractive southwest grew more slowly because of the paucity of horsecar routes. And in Oakland, California, real estate tycoon E. C. Sessions organized the Oakland Fruit Vale Railway Company in 1875 to service his newly subdivided Highland Park and Diamond districts. A second Oakland land baron, Walter Blair, founded the Broadway and Piedmont Railroad in 1876, with results that were noted by the *Oakland Daily Evening Tribune* on December 12, 1876:

An example of the value of a street railroad as an investment to improve or open up real estate can be seen in the Broadway and Piedmont Railroad which was completed last spring. This was a private enterprise projected by Walter Blair and Samuel and Montgomery Howe solely to give access to their landed property and to enhance the value of the property along the line. The increase in the value of property along the line as well as the high volume of sales which has accompanied the new line has proved the wisdom of the enterprise.⁵³

The way in which the new transit system altered the habits of the well-to-do almost immediately after its introduction is revealed in an 1859 diary entry of Sidney George Fisher, the scion of a wealthy Philadelphia family:

These passenger cars, as they are called, but which are street railroads with horse power, are a great convenience. Tho little more than a year old, they have almost displaced the heavy, jolting, slow and uncomfortable omnibus and are destined soon to banish it and hacks also entirely. They are roomy, their motion smooth and easy, they are clean, well cushioned and handsome, low to the ground so that it is convenient to get in or out and are driven at a rapid pace. They offer great facilities for traversing the city, now grown so large that the distances are very considerable from place to place. They traverse the city in its length and breadth and save time and expense. Today I took the 6th St. line at the turnpike and went to Shippen St., then walked to Elizabeth Fisher's wharf. Then, took the Pine St. line and went up to Mrs. Hone's in Pine above 19th St. Then took the Spruce St. line and came down to 10th St. and went from thence to Fisher's house. Coming home in the afternoon, I took the 5th St. line at Walnut St., close to Mr. Ingersoll's door and came out to the Germantown turnpike. Their remarkable success proves how much they were needed.⁵⁴

The privileged Fisher was hardly typical of working people, who had neither the leisure nor the income for such frequent crosstown trips. But the "riding habit" quickly caught on with the middle class, so much so that crowding became an early and persistent problem. A typical experience was related in a New York newspaper: "People are packed into them like sardines in a box, with perspiration for oil. The seats being more than filled, the passengers are placed in rows down the middle, where they hang on by the straps, like smoked hams in a corner grocery."⁵⁵

Nowhere was the horsecar impact more immediate or more profound than in the nation's largest city. In 1853, their first full year of operation, the New York horse railways carried about 7 million passengers. By 1856 they had laid twenty-three miles of track on little Manhattan Island, and both the Second and the Third Avenue lines had reached 60th Street. By 1860 track mileage had risen to 142 and ridership had reached 36 million, or about 100,000 per day.⁵⁶ By that time, the boundaries of a 45-minute commute reached the southern edge of Central Park, then under construction, and in periods of light traffic went beyond 80th Street. Although in 1856 the *Tribune* noted, "Out of town, which a few years ago meant above Canal Street, now means across the river or bay, far down by the seashore, or in the fast receding forests of adjoining counties," the practical impact of the horsecar was to provide a tremendous real-estate boost to upper Manhattan. By offering a comfortable means of land transportation, at least in comparison with available alternatives, the streetcar temporarily diverted the migration of taxpayers via ferry to Brooklyn and to New Jersey and insured the high-density residential coverage that would distinguish New York from other large cities.⁵⁷

Most importantly, the horsecars contributed to the development of the world's first integrated transportation systems. In New York, Philadelphia, Boston, and Chicago, they connected with the omnibuses which provided crosstown service, with the steam railroads which provided long-distance commuter service, and with the ferries which constantly trundled back and forth across the adjacent waterways. Many of the horsecar patrons were commuters on the railroads or ferries who used the horsecars for shorter trips from the terminals to their place of work. By the mid-1880s, there were 415 street railway companies in the United States operating over 6,000 miles of track and carrying 188 million passengers per year, or about 12 rides for every man, woman, and child who lived in a city of at least twenty-five hundred persons. The impact of such a ridership was in many respects foreseen by Fisher in 1859:

A beneficial effect of this will be to enable everyone to have a suburban or villa or country home, to spread the city over a vast space, with all the advantages of compactness and the advantages, moreover, of pure air, gardens, and rural pleasures. Before long, town life, life in close streets and alleys, will be confined to a few occupations, and cities will be mere collections of shops, warehouses, factories, and places of business.⁵⁸

Although the horsecar did not extend the building radius of the city nearly as much as the electric streetcar and the automobile would, Fisher's prediction was particularly appropriate for the United States, for it was there that the horsecar was most extensively and intensively adopted. As a contemporary summed up the impact: "It is hardly too much to say that the modern horse-car is among the most indispensable conditions of metropolitan growth. In these days of *fashionable effeminacy* and *flabby feebleness*, which never walks when it can possibly ride, the horsecar virtually fixes the ultimate limits of suburban growth."⁵⁹

Horsecar railways were built much more slowly in Europe, usually ten to fifteen years later than in North America. In 1869 only a few lines were operating on the continent, and as late as 1875 the total ridership of Paris, London, Vienna, and Berlin combined was much less than that for New York City alone. In Tokyo, the largest city in the Orient, the horsecar was not even introduced until 1882.⁶⁰

Comparisons with Queen Victoria's powerful and technologically advanced Great Britain are instructive. In Leeds the omnibus was the dominant mode of transit until the 1890s, a generation after those clumsy wagons had been mostly withdrawn from the streets of American cities. Similarly, the horsecar itself was to remain in active English service until World War I, long after the electric trolley had replaced the ubiquitous horse in United States streetcar systems. This rapid American exploitation of new means of transport was a reflection of a national temperament which emphasized mobility and change. In Great Britain, by contrast, no less a personage than the Duke of Wellington, the victor over Napoleon at Waterloo, thought it a mistake to build railroads because they would "only encourage the common people to move about needlessly."⁶¹

Transportation Innovation and Suburban Growth

Transportation change is not a sufficient explanation for the initial development of the suburban trend. If that were the case then every great

city that shared the wealth of the Industrial Revolution and the technology of the horsecar and the railroad would have exhibited a similar residential pattern. To be sure, as Adna Ferrin Weber noted in his exhaustive *The Growth of Cities in the Nineteenth Century*, metropolitan growth on every continent was most rapid on the fringes. In London a new environment was taking shape in Barnes, Hampstead, Putney, Hammer-smith, and St. John's Wood by the end of the eighteenth century, and by the 1840s the vastness of these suburbs was filling visitors with wonder. Utilizing both private carriages and public omnibuses, the aristocracy was transforming the small villages to the west into fashionable places of residence. In 1849 Thomas Babington Macaulay reported that the "chiefs of the mercantile interests" had already removed their families to "suburban country seats surrounded by shrubberies and flower gardens. . . . Lombard Street and Threadneedle Street are merely places where men toil and accumulate. They go elsewhere to enjoy and to expend." In fact, the great mansions and landscaped parks of "Imperial Kensington" were so influential that those citizens of London interested in social status were finding it necessary to remove themselves to the "court suburb" by midcentury. As in the United States, the upper classes were the first, and the working classes the last to move into the commuting suburbs. Even today London remains a city of low population density—a collection of villages anxious to avoid the type of concentration typical of the rest of Europe.⁶²

This pattern was hardly inevitable. The privileged groups in large American and British cities could have retained their convenient domiciles in the core and left the shabby periphery to the poor. This is what happened in Europe, Asia, and South America, where new neighborhoods were typically densely settled and inhabited by the less fortunate, rather than the well-to-do. And even these poorer denizens crowded as close to the city as they could get. One mile beyond Paris in 1850 placed one in a wilderness of gypsum quarries and bond hills; at a similar distance from the walls of Rome the traveler was quite literally in an immense expanse of emptiness. In the 1890s, Weber calculated that the population density of fifteen American cities averaged twenty-two people per acre as compared with 157.6 for thirteen German cities, and he speculated about the phenomenon:⁶³

It has sometimes been urged that this is largely the result of the development of the electric street railway in America, but the causal connection is not apparent. . . . It should rather be said that the American penchant for dwelling in cottage homes instead of business blocks after the fashion of Europe is the cause, and the trolley car the effect.

For the underlying causes of the increasingly stratified and segregated social geography of great American cities, as well as their relatively low density as compared to Europe, we must look not just to transportation technology and the powerful mechanical forces unleashed by the Industrial Revolution but to the development of new cultural values.